

EXERCISE B

1) INPUT: POLICY π , MDP
OUTPUT: V_π (AN ESTIMATION OF)

- INITIALIZE $V(s) \forall s \in \mathcal{S}$ ARBITRARILY
RETURNS $(s) \leftarrow \text{EMPTY}$ FOR ALL $s \in \mathcal{S}$

- LOOP FOR EACH EPISODE

GENERATE AN EPISODE FOLLOWING π :

$s_0, a_0, r_0, s_1, a_1, r_2, \dots, s_{T-1}, a_{T-1}, r_T$

$G \leftarrow 0$

LOOP FOR EACH STEP OF THE EPISODE
GOING BACKWARDS $t = T-1, T-2, \dots, 0$

$G \leftarrow \gamma G + R_{t+1}$

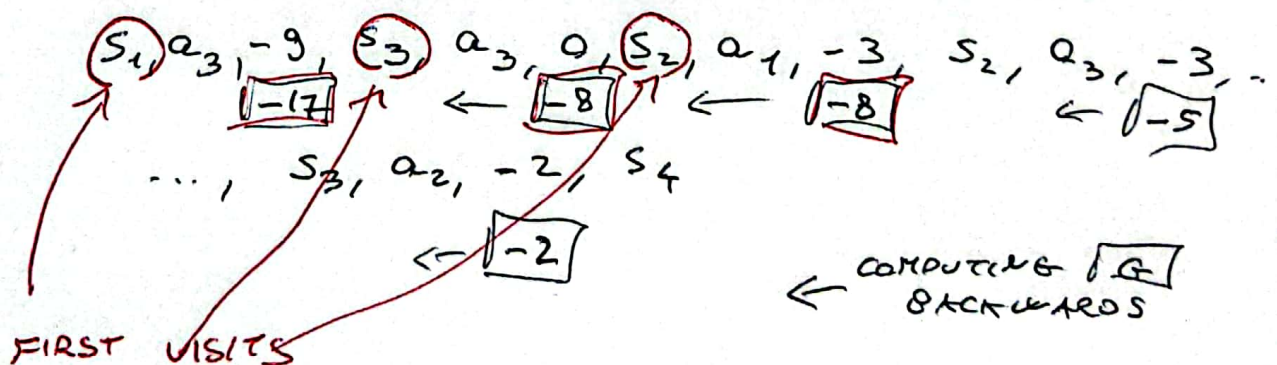
UNLESS s_t APPEARS IN $s_0, \dots, s_{t-1}$

APPEND G TO RETURNS (s_t)

$V(s_t) \leftarrow \text{AVERAGE}(\text{RETURNS}(s_t))$

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2) EPISODE #1:

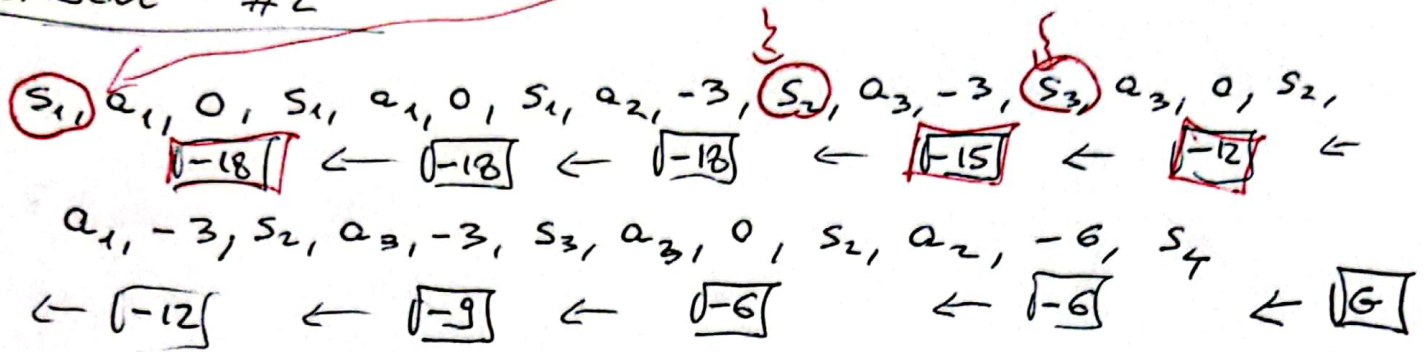


$G_1: -17$

$G_2: -8$

$G_3: -8$

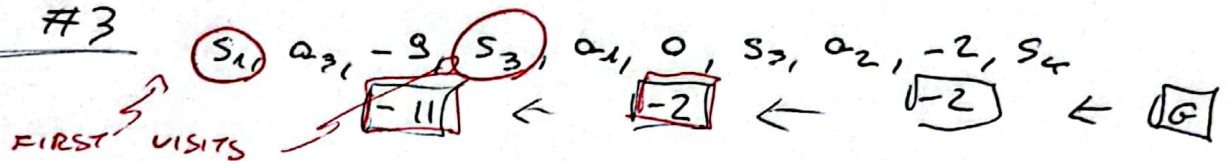
FIRST VISITS



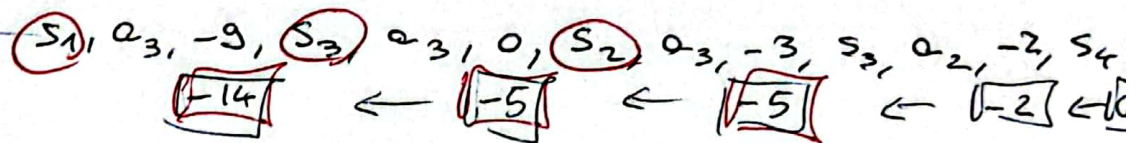
$G_1! - 18$

$$G_2: -15$$
$$G_3: -12$$

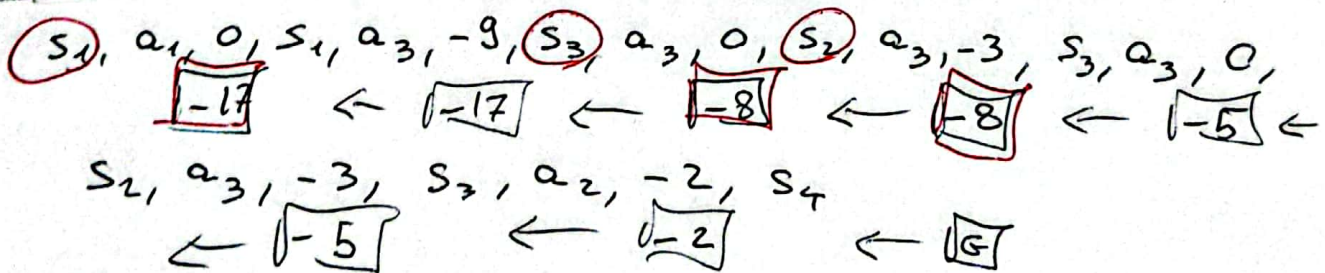
EPISODE #3


$$G_1: -11; \quad G_2: /, \quad G_3: -2$$

EPISODE # 4


$$G_1: -14, \quad G_2: -5, \quad G_3: -5$$

EPISODE # 5


$$G_1: -17, \quad G_2: -8, \quad G_3: -8$$

$$V_T(s_1) = - \frac{17+18+11+14+17}{5} = -15.4$$

$$V_{\pi}(s_2) = - \frac{8+15+5+8}{4} = -9$$

$$V_{\pi}(S_3) = - \frac{8+12+2+5+8}{5} = -7$$

V_{π} NOT FAR FROM THE TRUE $V_{\pi} =$

3) THE CHANGES IN THE ALGORITHM ARE THE FOLLOWING:

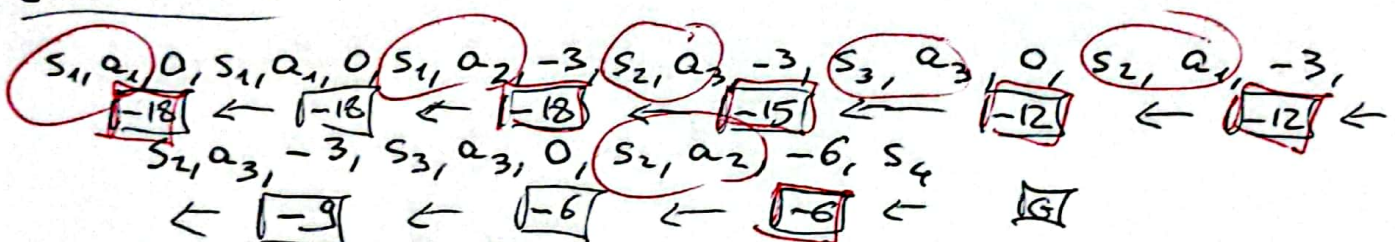
- $Q(s, a)$ A MATRIX OF SIZE $|S|, |A|$ IS CONSIDERED INSTEAD OF $V(s)$
- VERIFICATION ON THE LAST LOOP IS 'UNLESS (s_t, a_t) APPEARS IN $(s_0, a_0), \dots, (s_{t-1}, a_{t-1})$ '

4) EPISODE #1



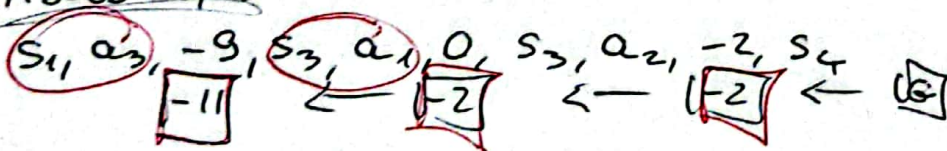
$$G(s, a) = \begin{matrix} & a_1 & a_2 & a_3 \\ \begin{matrix} s_1 \\ s_2 \\ s_3 \end{matrix} & \begin{bmatrix} / & / & -17 \\ -8 & / & -5 \\ / & -2 & -8 \end{bmatrix} \end{matrix}$$

EPISODE #2



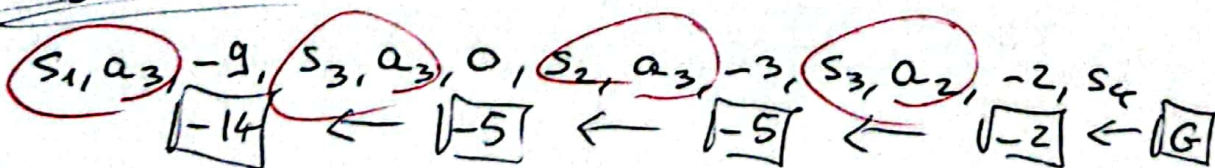
$$G(s, a) = \begin{matrix} & a_1 & a_2 & a_3 \\ \begin{matrix} s_1 \\ s_2 \\ s_3 \end{matrix} & \begin{bmatrix} -18 & -18 & / \\ -12 & -6 & -15 \\ / & / & -12 \end{bmatrix} \end{matrix}$$

EPISODE #3



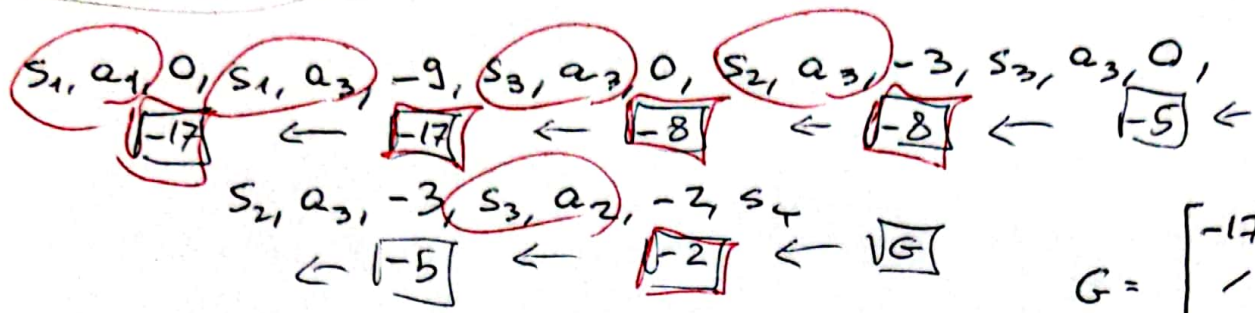
$$G = \begin{bmatrix} / & / & -11 \\ / & / & / \\ -2 & -2 & / \end{bmatrix}$$

EPISODE #4



$$G = \begin{bmatrix} / & / & -14 \\ / & / & -5 \\ / & -2 & -5 \end{bmatrix}$$

EPISODE #5



$$G = \begin{bmatrix} -17 & -17 \\ / & / & -8 \\ / & -2 & -8 \end{bmatrix}$$

$$Q_{\pi}(s, a) = \begin{bmatrix} \frac{-18-17}{2} & -18 & \frac{-17-11-14-12}{4} \\ \frac{-8-12}{2} & -6 & \frac{-5-15-5-8}{4} \\ -2 & \frac{-2-2-2-2}{4} & \frac{-8-12-5-8}{4} \end{bmatrix} = \begin{bmatrix} -17.5 & -18 & -14.75 \\ -10 & -6 & -8.25 \\ -2 & -2 & -8.25 \end{bmatrix} \quad (*)$$

5) WITHOUT KNOWLEDGE OF THE MDP, WE CANNOT USE V_{π} FOR DERIVING A NEW POLICY π' : WE CAN HOWEVER EXPLOIT Q_{π}

BASED ON $Q_{\pi} (*)$ WE CAN ACT GREEDILY AND CHOOSE!

$$\pi'(s_1) = a_3$$

$$\pi'(s_2) = a_2$$

$$\pi'(s_3) = a_1 \text{ AND } a_2 \text{ WITH EQUAL PROBABILITY (ARBITRARY CHOICE)}$$

EXPLORING STARTS OR ϵ -SOFT MODIFICATIONS CAN BE INTRODUCED TO AVOID EXPLOITATION PROBLEMS.

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